

● HARD

● ADVANCED MATH

● ANSWER KEY

Advanced Math

17

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Q1**A**

A. $0 = x^2 - 24x - 2,661$

Choice A is correct. It's given that the banner's length x , in inches, is 24 inches longer than its width, in inches. It follows that the banner's width, in inches, can be represented by the expression $x - 24$. The area of a rectangle is the product of its length and its width. It's given that the area of the banner is 2,661 square inches, so it follows that $2,661 = xx - 24$, or $2,661 = x^2 - 24x$. Subtracting 2,661 from each side of this equation yields $0 = x^2 - 24x - 2,661$. Therefore, the equation that represents this situation is $0 = x^2 - 24x - 2,661$.

Choice B is incorrect and may result from representing the width, in inches, of the banner as $24 - x$, rather than $x - 24$.

Choice C is incorrect and may result from representing the width, in inches, of the banner as $x + 24$, rather than $x - 24$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q2

B

B. $3 + 2\frac{55^2}{4}$

Choice B is correct. It's given that function g is defined by $g(t) = f(t) + 3$ and that $f(t) = 55t - 2t^2$. Substituting $55t - 2t^2$ for $f(t)$ in $g(t) = f(t) + 3$ yields $g(t) = 55t - 2t^2 + 3$, or $g(t) = -2t^2 + 55t + 3$. The maximum value of $g(t)$ can be found by completing the square to rewrite the equation defining g in the form $g(t) = a(t - h)^2 + k$, where the maximum value of the function is k , which occurs when $t = h$, and a is a negative constant. The equation $g(t) = -2t^2 + 55t + 3$ is equivalent to $g(t) = -2\left(t^2 - \frac{55}{2}t\right) + 3$, which can be rewritten as $g(t) = -2\left(t^2 - \frac{55}{2}t + \left(\frac{55}{4}\right)^2\right) + 3 + 2\left(\frac{55}{4}\right)^2$, or $g(t) = -2\left(t - \frac{55}{4}\right)^2 + 3 + 2\left(\frac{55}{4}\right)^2$. This equation is in the form $g(t) = a(t - h)^2 + k$, where $a = -2$, $h = \frac{55}{4}$, and $k = 3 + 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Alternate approach: Since the function f is a quadratic function, the maximum value of $f(t)$ occurs at the value of t that's halfway between the two zeros of the function. The zeros of function f can be found by substituting 0 for $f(t)$ in the equation defining f , which yields $0 = 55t - 2t^2$. This equation can be rewritten as $0 = t(55 - 2t)$. By the zero product property, it follows that $t = 0$ or $55 - 2t = 0$. Subtracting 55 from each side of the equation $55 - 2t = 0$ yields $-2t = -55$. Dividing each side of this equation by -2 yields $t = \frac{55}{2}$. Therefore, the zeros of function f are 0 and $\frac{55}{2}$. The value that's halfway between 0 and $\frac{55}{2}$ can be found by calculating the average of 0 and $\frac{55}{2}$, which is $\frac{0 + \frac{55}{2}}{2}$, or $\frac{55}{4}$. It follows that the maximum of function f occurs when $t = \frac{55}{4}$. Substituting $\frac{55}{4}$ for t in the equation defining function f yields $f\left(\frac{55}{4}\right) = 55\left(\frac{55}{4}\right) - 2\left(\frac{55}{4}\right)^2$, which is equivalent to $f\left(\frac{55}{4}\right) = \frac{55^2}{4} - 2\left(\frac{55^2}{4^2}\right)$. Multiplying $\frac{55^2}{4}$ by $\frac{4}{4}$ in this equation to get a common denominator yields $f\left(\frac{55}{4}\right) = 4\left(\frac{55^2}{4^2}\right) - 2\left(\frac{55^2}{4^2}\right)$, or $f\left(\frac{55}{4}\right) = 2\left(\frac{55^2}{4^2}\right)$, which is equivalent to $f\left(\frac{55}{4}\right) = 2\left(\frac{55}{4}\right)^2$. Thus, the maximum value of $f(t)$ is $2\left(\frac{55}{4}\right)^2$. Since the equation defining $g(t)$ is $g(t) = f(t) + 3$, the maximum value of $g(t)$ is 3 greater than the maximum value of $f(t)$. It follows that the maximum value of $g(t)$ is $3 + 2\left(\frac{55}{4}\right)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3**D**

D. Neither I nor II

Choice D is correct. A function of the form $f(x) = ab^x + c$, where $a < 0$ and $b > 1$, is a decreasing function. Both of the given functions are of this form; therefore, both are decreasing functions. If a function f is decreasing as the value of x increases, the corresponding value of $f(x)$ decreases; therefore, the function doesn't have a minimum value. Thus, neither of the given functions has a minimum value.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Q4**8**

The correct answer is 8. The values of $f(x)$ for the exponential function f shown in the table increase by a factor of a^4 for each increase of 1 in x . This relationship can be represented by the equation $f(x) = a^{4x+b}$, where b is a constant. It's given that when $x = 2$, $f(x) = a^5$.

Substituting 2 for x and a^5 for $f(x)$ into $f(x) = a^{4x+b}$ yields $a^5 = a^{4(2)+b}$. Since $4(2) + b = 5$, it follows that $b = -3$. Thus, an equation that defines the function f is $f(x) = a^{4x-3}$. It follows that the value of k such that $f(k) = a^{29}$ can be found by solving the equation $4k - 3 = 29$, which yields $k = 8$.

Q5**A****A.** 16

Choice A is correct. The equation $fx = 1.84^{\frac{x}{4}}$ can be rewritten as $fx = 1.84^{\frac{1}{4}x}$, which is equivalent to $fx = 1.84^{\frac{1}{4}x}$, or approximately $fx = 1.16467^x$. Since it's given that $fx = 1.84^{\frac{x}{4}}$ can be rewritten as $fx = 1 + \frac{p}{100}^x$, where p is a constant, it follows that $1 + \frac{p}{100}$ is approximately equal to 1.16467. Therefore, $\frac{p}{100}$ is approximately equal to 0.16467. It follows that the value of p is approximately equal to 16.467. Of the given choices, 16 is closest to the value of p .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q6**A****A.** $M = 1,800(1.02)^{\frac{q}{4}}$

Choice A is correct. In 1 year, there are 4 quarter years, so the number of quarter years, q , is 4 times the number of years, t ; that is, $q = 4t$. This is equivalent to $t = \frac{q}{4}$, and substituting this into the expression for M in terms of t gives $M = 1,800(1.02)^{\frac{q}{4}}$.

Choices B and D are incorrect and may be the result of incorrectly using $t = 4q$ instead of $q = 4t$. (Choices B and D are nearly the same since 1.02^{4q} is equivalent to $(1.02^4)^q$, which is approximately 1.082^q .) Choice C is incorrect and may be the result of incorrectly using $t = 4q$ and unnecessarily dividing 0.02 by 4.

Q7**-19**

The correct answer is -19. It's given that function f is defined by $fx = a^x + b$, where a and b are constants and $a > 0$. It's also given that the graph of $y = fx$ in the xy -plane has a y -intercept at 0, -25 and passes through the point 2, 23. Since the graph has a y -intercept at 0, -25, $f0 = -25$. Substituting 0 for x in the given equation yields $f0 = a^0 + b$, or $f0 = 1 + b$, and substituting -25 for $f0$ in this equation yields $-25 = 1 + b$. Subtracting 1 from each side of this equation yields $-26 = b$. Substituting -26 for b in the equation $fx = a^x + b$ yields $fx = a^x - 26$. Since the graph also passes through the point 2, 23, $f2 = 23$. Substituting 2 for x in the equation $fx = a^x - 26$ yields $f2 = a^2 - 26$, and substituting 23 for $f2$ yields $23 = a^2 - 26$. Adding 26 to each side of this equation yields $49 = a^2$. Taking the square root of both sides of this equation yields $\pm 7 = a$. Since it's given that $a > 0$, the value of a is 7. It follows that the value of $a + b$ is $7 - 26$, or -19.

Q8**A****A.** $hx = 172^x$

Choice A is correct. It's given that $f(x) = 272(2)^x$ and $h(x) = f(x - 4)$. Substituting $x - 4$ for x in $f(x) = 272(2)^x$ yields $fx - 4 = 2722^{x-4}$. Substituting hx for $fx - 4$ in this equation yields $hx = 2722^{x-4}$. Using the properties of exponents, the function $h(x) = 272(2)^{x-4}$ can be rewritten as $hx = \frac{2722^x}{2^4}$, which is equivalent to $hx = \frac{2722^x}{16}$, or $h(x) = 17(2)^x$. Therefore, of the given choices, an equation that defines function h is $h(x) = 17(2)^x$.

Choice B is incorrect. This equation defines function h if $h(x) = f(x - 2)$, not $h(x) = f(x - 4)$.

Choice C is incorrect. This equation defines function h if $h(x) = f(4x)$, not $h(x) = f(x - 4)$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9

D

D. $-\frac{3}{2}$

Choice D is correct. It's given that $f(x) = x^2 - 10x + 13$, which can be rewritten as $f(x) = x^2 + 3x - 130$. Since the coefficient of the x^2 -term is positive, the graph of $y = f(x)$ in the xy -plane opens upward and reaches its minimum value at its vertex. The x -coordinate of the vertex is the value of x such that $f(x)$ reaches its minimum. For an equation in the form $f(x) = ax^2 + bx + c$, where a , b , and c are constants, the x -coordinate of the vertex is $-\frac{b}{2a}$. For the equation $f(x) = x^2 + 3x - 130$, $a = 1$, $b = 3$, and $c = -130$. It follows that the x -coordinate of the vertex is $-\frac{3}{2}$, or $-\frac{3}{2}$. Therefore, $f(x)$ reaches its minimum when the value of x is $-\frac{3}{2}$.

Alternate approach: The value of x for the vertex of a parabola is the x -value of the midpoint between the two x -intercepts of the parabola. Since it's given that $f(x) = x^2 - 10x + 13$, it follows that the two x -intercepts of the graph of $y = f(x)$ in the xy -plane occur when $x = 10$ and $x = -13$, or at the points $(10, 0)$ and $(-13, 0)$. The midpoint between two points, (x_1, y_1) and (x_2, y_2) , is $\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$. Therefore, the midpoint between $(10, 0)$ and $(-13, 0)$ is $\left(\frac{10 + (-13)}{2}, \frac{0 + 0}{2}\right)$, or $\left(-\frac{3}{2}, 0\right)$. It follows that $f(x)$ reaches its minimum when the value of x is $-\frac{3}{2}$.

Choice A is incorrect. This is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane.

Choice B is incorrect. This is one of the x -coordinates of the x -intercepts of the graph of $y = f(x)$ in the xy -plane.

Choice C is incorrect and may result from conceptual or calculation errors.

Q10

C

C. The unit price, in dollars, of the product that will result in a revenue of \$0

Choice C is correct. By definition, the y-value when a function intersects the x-axis is 0. It's given that the graph of the function intersects the x-axis at 0 and a, that x is the unit price, in dollars, of a product, and that $f(x)$, where $y = f(x)$, is the revenue, in dollars, that a company receives from the sales of the product. Since the value of a occurs when $y = 0$, a is the unit price, in dollars, of the product that will result in a revenue of \$0.

Choice A is incorrect. The revenue, in dollars, when the unit price of the product is \$0 is represented by $f(x)$, when $x = 0$. Choice B is incorrect. The unit price, in dollars, of the product that will result in maximum revenue is represented by the x-coordinate of the maximum of f. Choice D is incorrect. The maximum revenue, in dollars, that the company can make is represented by the y-coordinate of the maximum of f.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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